

Response to Referee R6

We thank the editor and referees for their continued engagement with our paper. Below we provide a brief summary of the main changes in this revision and then respond to each comment in turn.

- **Stock-return analysis (R2).** R2 asked for a fuller treatment of the stock market evidence around a greater set of legal events. The examination the referee suggested identified additional important events: the certiorari grant and midterm election of November 1934 (the certiorari grant came on November 5 and the election on November 6, a day on which the market was closed, so their individual contributions cannot be distinguished). Over this window, gold-exposed firms outperformed non-exposed firms by 3.5 percentage points ($t = 3.3$), second only to the Joint Resolution window. The event is now part of the main analysis. The return evidence appears in a dedicated table (Table 1, in the main text) and four event-study figures (Internet Appendix Figures IA.1–IA.4). A new Internet Appendix section examines the lower court events; none of them generated a strong differential in returns between exposed and unexposed firms.
- **Robustness with the full set of fixed effects (R6).** R6 asked to see the characteristic-control specifications with industry-year fixed effects included. New Internet Appendix Table IA.16 reports all of them. The 1933 effect is essentially unchanged and remains significant; the 1934 effect attenuates and mostly loses significance, without sign reversals. Section 5 now discusses this sensitivity directly and notes that the attenuation arises only when the characteristic controls and the industry-year fixed effects are included together.
- **Limitations of the exposure measure (R6).** R6 asked the paper to be more upfront about the fact that gold exposure is entangled with the debt-versus-preferred composition of a firm’s fixed claims. Section 5.5 now closes with an explicit limitations discussion, stating that our placebo and control strategies mitigate but cannot fully eliminate this concern.
- **Pre-trends (R6).** R6 noted that the litigation-year coefficients cannot be statistically distin-

guished from individual pre-treatment estimates such as the 1930 coefficient. Section 5.2 now acknowledges this directly and clarifies that the support for the parallel-trends assumption comes from the pre-treatment coefficients as a group: individually insignificant and without trend toward the treatment years.

- **Liberty bond discussion (R2).** R2 asked for the appropriate context for the Liberty Bond price increase during the January 1935 arguments. We agree with the referee’s interpretation and have adopted it in the text: as gold-denominated government obligations, Liberty Bonds rose in price because investors saw a real chance the government could lose the case. The missing *New York Times* citation has been added.

Corrections identified while reviewing the code. As part of this revision we reviewed the code of our empirical exercises end to end; this led to a few corrections, none of which changed the interpretation of the results.

1. The “no redemption” sample of column 4 of Table 4 (and of Internet Appendix Table IA.9, column 1) inadvertently excluded 37 unexposed firms that lacked a prior-year observation during 1931–1935. The revised sample excludes only the firms that retired their gold-clause debt during the litigation window. No coefficient loses significance; the level of $\tilde{d}$ and the $1935 \times \tilde{d}$ interaction sharpen from the 5% to the 1% level. For convenience, Table R1, at the end of this document, reports column 4 before and after the correction.
2. The standard errors in column 7 of Table 4 (Table 3 in the previous draft), the bank-debt placebo, were inadvertently clustered by firm only in the previous draft; they are now two-way clustered by firm and year, like every other column. The coefficients are unchanged; the 1926 and 1928 placebo interactions sharpen from the 10% level to the 5% and 1% levels, respectively, and the litigation-year cells remain insignificant in both versions. For convenience, Table R2, at the end of this document, reports the column before and after the correction.
3. In Internet Appendix Table IA.14, the below/above-median classification of firms was inadvertently computed over all sample years rather than at the 1930 base year. The cor-

rected classification changes the estimates, while the central heterogeneity conclusions are unchanged.

4. The firm count printed in the header of Panel C of Table 2 (formerly 594) was a typographical error introduced when the table was created; the underlying sample is unchanged (503 firms).

First, I appreciate the improvement in measuring d_{it} relative to all long-term obligations involving fixed contractual payments, but this only partly addresses my concerns. Preferred shares are not interest bearing, and thus may behave quite differently than debt during downturns or when firms are close to distress. I understand that there is nothing the author can do about this because bank debt is simply too small, but the paper should acknowledge this limitation more clearly and briefly discuss the potential confounders that cannot be addressed due to how firms structured their debt at that time.

The referee raises a valid point, and we now address it explicitly in the manuscript. Preferred dividends are discretionary in a way that bond coupons are not—deferring them does not trigger default—and we agree that this distinction matters in distress, even though the flexibility was partial, as preferred issues of the era were typically cumulative. Because preferred shares enter the denominator of $\tilde{d}$ but not the numerator, the concern is that high- $\tilde{d}$ firms, which rely more on inflexible bond financing, may be more vulnerable to downturns generally, so that the 1933–1934 differential partly reflects debt rigidity rather than gold clause litigation.

The evidence that speaks most directly to this concern is the temporal pattern (Figure 3): the effect is concentrated in 1933–1934 and reverses after the 1935 ruling, which is difficult to reconcile with debt rigidity alone, as that mechanism would also operate in the 1937–1938 recession, where we find no differential effect. The industry-year fixed effects and characteristic-interacted controls of Table 7 provide further protection against shocks correlated with industry and observable firm structure. The preferred-share placebo (column 6 of Table 4), in contrast, does not cleanly distinguish gold-clause exposure from fixed-claim composition: its positive but imprecise coefficients are compatible with preferred-heavy firms having more flexible payment obligations, and we should not treat the lack of statistical significance as dispositive evidence against the composition concern.

Section 5.5 now closes with an explicit limitations paragraph; it reads in part:

“[B]ecause virtually all corporate bonds of the era carried gold clauses while preferred shares carried none, $\tilde{d}$ is entangled with the composition of a firm’s fixed claims: preferred shares enter its denominator but not its numerator, so preferred-heavy firms have

mechanically lower measured exposure. Because preferred dividends are discretionary and non-interest-bearing (although the flexibility was partial, as preferred issues of the era were typically cumulative), such firms may also behave differently in distress, raising the concern that $\tilde{d}$ partly captures a general sensitivity to downturns among bond-reliant firms ... An ideal design would vary gold-clause exposure while holding the debt-versus-preferred composition fixed; the institutional structure of the period precludes this.”

I appreciate using 1936–37 as a placebo. While this helps address concerns about downturns driving the results, there are other important macro differences between both periods—such as the collapse of bond issuance, that limited rollover—that may affect exposed firms more in 1933–34 beyond the uncertainty of the abrogation of the gold clause. That is, the placebo helps but only for a subset of the concerns about confounders. To that end, the robustness regressions are more helpful. Though the findings are broadly reassuring, they do point to either the thinness of the data or a lack of robustness. The authors cannot control for industry $\times$ year and $X_{it} \times$ year at the same time, so the fixed effects vary across specifications in Table 6. The paper argues this is done “To assess robustness to firm-level controls separately from industry-level controls, this specification omits industry-year fixed effects.” This sounds like a copout. Given the concerns of New Deal policies as confounders, which may have affected firms differently given their characteristics, one may want to control for both. My sense is that they lack power, as even without the industry-year FEs the significance is lost for $1934 \times d$ across several specifications for Columns 2–10. I would like to see the results with industry-year FEs for these columns as well. If significance is lost across the board, then the paper should acknowledge it and make its best case to defend their preferred specifications due to limited power—but readers should know whether the effects are sensitive so they can pass their own judgement.

We thank the referee for these comments and address each in turn.

Placebo and the rollover channel. The referee correctly notes that the 1936–37 placebo addresses recession-driven confounders but not the collapse of bond issuance that may have disproportionately

affected gold-exposed firms—which held more bonds—through a rollover channel (the collapse in industrial bond issuance is shown in Figure 4 of the paper). Column 3 of Table 4 speaks to this concern without fully resolving it: it excludes firms with bonds maturing at any point between 1931 and 1934, and the 1933 and 1934 coefficients remain -0.058 and -0.037 , essentially unchanged from the baseline, suggesting that the investment effect is not entirely driven by firms with immediate refinancing needs. We also say this in the paper now: Section 5.2 notes that “[t]his placebo . . . addresses only a subset of the possible confounders. Important macroeconomic differences between the two periods, such as the collapse of bond issuance and the limits it placed on rollover, may have affected exposed firms in 1933–1934 beyond the uncertainty of the abrogation. The sample-restriction tests below and the further robustness checks in Section 5.5 speak to this concern.”

Industry-year fixed effects in columns 2–10. We agree that readers should be able to assess this sensitivity directly. Internet Appendix Table IA.16 re-estimates columns 2–10 of Table 7 (Table 6 in the previous draft) with industry-year fixed effects in place of year fixed effects, so that all ten specifications share the fixed-effect structure of column 1; its columns (1)–(9) correspond to columns 2–10 of Table 7. For convenience, we reproduce it as Table R3 at the end of this document.

The referee asked whether the effects become noisy or whether they become small and insignificant. The answer differs by year, and Section 5 now states it in exactly these terms. For $1933 \times \tilde{d}$, the estimates are essentially unchanged and remain significant—at the 5% level in seven of nine specifications and at the 10% level in all nine. For $1934 \times \tilde{d}$, significance is lost in seven of nine specifications, and the loss comes mainly from attenuation of the point estimates (which range from -0.012 to -0.031 , against -0.022 to -0.043 under year fixed effects) rather than from wider standard errors. The estimates never reverse sign, and the attenuation appears only when characteristic-period controls are stacked on top of industry-year cells: column 1 of Table 7, which absorbs industry-year fixed effects without characteristic controls, estimates $1934 \times \tilde{d}$ at -0.036 , significant at the 5% level.

The characteristic-period interactions and the industry-year cells absorb overlapping variation. The attenuation of the 1934 estimates comes through smaller point estimates rather than wider standard

errors.

Column 1 of Table 7 remains our preferred specification for the New Deal confounder concern: industry-year fixed effects absorb any shock common to an industry in a given year. The characteristic controls address the distinct concern that gold exposure proxies for observables on differential investment trajectories, and they continue to do so under industry-year fixed effects. Combining the two exercises attenuates the 1934 estimate; we say so explicitly in Section 5 and provide Table IA.16 so that readers can judge the sensitivity themselves.

The authors made no comment about the fact that the 1933 and 1934 coefficients do not seem statistically different than the 1930 effects (Figure 3).

The referee is right: the 1933 and 1934 coefficients cannot be statistically distinguished from the 1930 coefficient. We now acknowledge this directly in the paper; Section 5.2 adds:

“We acknowledge that, given the width of the confidence intervals around individual year estimates, the 1933 and 1934 coefficients cannot be statistically distinguished from individual pre-treatment estimates such as the 1930 coefficient; the support for the parallel-trends assumption comes instead from the pre-treatment coefficients as a group, which are individually insignificant and show no trend toward the treatment years.”

That pattern—pre-treatment coefficients that are uniformly small and individually insignificant, with 1931, the last estimated pre-treatment year, essentially at zero relative to the omitted 1932 base—is what the design relies on.

One small comment—my understanding (though I could be wrong) is that repurchases in the open market were not allowed post 1929. I imagine the main reason driving the decline in shares was reorganizations. If that is the case, it is not obvious one would want to include this if the point of the analysis is to estimate whether firms were using payout policy to preemptively give the cash to shareholders. That said, the results in Table 4 show that the effects are driven by dividends, not repurchases, so this is mostly about guiding modern readers on what firm policies were in the 1930s.

The referee’s comment prompted us to examine this directly in our data. Two facts stand out. First, treasury stock does appear on the Moody’s balance sheets in our sample: between a fifth and a third of firms report a nonzero treasury-stock position during 1930–1934, and net additions to treasury stock occur in roughly one in eight firm-years. We cannot tell from the balance sheet how these shares were acquired—open-market purchases, private transactions, exchanges, or settlements—so this observation does not by itself settle the institutional question the referee raises. Whatever their source, the amounts are small: the median net addition is about 0.5% of total assets. Second, our net repurchase measure is built from split-adjusted changes in shares outstanding, and over 1930–1934 the number of shares outstanding is typically unchanged (after accounting for splits), with most of the remaining variation coming from net issuance rather than share reductions. The referee’s economic point therefore stands regardless of the institutional details: nothing in this period resembles the discretionary buyback programs of modern data, and net repurchases were not a meaningful payout vehicle.

We agree that modern readers would benefit from this context, and we have added a footnote to Section 5 stating these facts: treasury-stock additions occur but are small, the number of shares outstanding is typically unchanged (after accounting for splits), and the variation in the net repurchase measure comes mostly from share issuance rather than retirement.

This does not affect any of our substantive conclusions. As the referee notes, Table 5 (Table 4 in the previous draft) shows that the payout effect is driven entirely by cash dividends (column 2 is significant), while the net repurchase coefficient (column 3) is statistically insignificant in 1933 and 1934. The channel relevant to our debt overhang interpretation—shareholders extracting value through discretionary distributions in anticipation of losing their claim on the assets—operates through dividends, not through changes in shares outstanding.

Additional comment (cover letter)

A key issue is whether the authors can clearly disentangle the effects of uncertainty about the abrogation of gold clauses from other confounders—other policies and firm characteristics, and whether it is about the gold clause vs interest-bearing leverage. There is not much they can do about

the second given how firms structured their debt. I appreciate the refinement in the definition of the treatment effect, but the ideal case would be not to have preferred shares either. I am willing to look passed this, but the paper should be more upfront. The bigger issue is other confounders. They have tried to address this in the robustness section in Table 6, but as you can see they cannot control for a full set of fixed effects. What bothers me more is that they have not shown us the results with all effects, and instead state they do not include industry-year FEs in all columns to assess the robustness to industry and firm separately. This is a weak argument. If the issue is that they have too few firms to do this without the treatment effects being killed, they should say that. Readers may understand, but they need to know. And I would like to see how bad effects get—is it noisy or do the results become small and insignificant? I also find the placebo less helpful than the paper claims, but that’s a personal taste.

We are grateful for the referee’s overall assessment and for the clarity of this summary. We agree with the central message—that the paper should be upfront about what the data can and cannot establish—and we have revised the manuscript accordingly. We respond to the two main issues in turn, and indicate where each is now addressed in the paper.

Gold clause exposure versus interest-bearing leverage (preferred shares). The referee is correct that the ideal design would isolate gold-clause exposure while holding the debt-versus-preferred composition fixed, and that the institutional structure of the period does not permit this: virtually all corporate bonds of the era carried gold clauses, while preferred shares carried none. We no longer leave this implicit. Section 5.5 now contains an explicit limitations paragraph stating that $\tilde{d}$ is necessarily entangled with the composition of a firm’s fixed claims, that preferred-heavy firms may behave differently in distress for reasons unrelated to gold clauses, and that our tests mitigate but cannot fully eliminate this concern.

Other confounders and the full set of fixed effects. We agree that the previous draft’s justification for not always including industry-year fixed effects read as a rationalization, and we have removed that framing. We now do two things. First, we state the real reason plainly in Section 5.5: with a sample of this size, absorbing industry-year variation and characteristic-period variation simultaneously

exhausts the identifying variation available to estimate the treatment effects. Second, and most importantly, we now show the results. Internet Appendix Table IA.16 re-estimates all of columns 2–10 of Table 7 with industry-year fixed effects, so the reader sees exactly “how bad effects get.” The answer differs by year. For $1933 \times \tilde{d}$, the estimates are essentially unchanged and remain significant—at the 5% level in seven of nine specifications and at the 10% level in all nine. For $1934 \times \tilde{d}$, significance is lost in seven of nine specifications, and the loss comes mainly from attenuation of the point estimates (which range from -0.012 to -0.031 , against -0.022 to -0.043 under year fixed effects) rather than from wider standard errors. The estimates never reverse sign, and the attenuation appears only when characteristic-period controls are stacked on top of industry-year cells: column 1 of Table 7, which absorbs industry-year fixed effects without characteristic controls, estimates $1934 \times \tilde{d}$ at -0.036 , significant at the 5% level. The effect does not reverse or vanish, and we now say so directly rather than asking the reader to take it on faith.

On the placebo. We take the referee’s point that the 1936–37 placebo addresses only a subset of confounders—chiefly recession-driven ones—and not, for example, the collapse of bond issuance (Figure 4 of the paper) and the attendant rollover risk. We have tempered the paper’s language about the placebo accordingly and now lean more heavily on the direct robustness checks, including the exclusion of firms with bonds maturing during the litigation window (column 3 of Table 4), which speaks specifically to the rollover channel.

Taken together, these revisions are intended to make the paper’s limitations visible to readers while preserving what we believe is the core, robust finding: gold-exposed firms reduced investment specifically during the 1933–1934 litigation window and reversed afterward, a pattern that survives industry-year fixed effects in our preferred specification and that is not reproduced by the preferred-share or bank-debt placebos.

Table R1: Table 4, column 4 (no-redemption sample), before and after the correction

	Column 4 (No redemption)	
	Round 1	Revised
Q	0.085*** (0.015)	0.085*** (0.014)
$\tilde{d}$	-0.097** (0.033)	-0.099*** (0.033)
$1926 \times \tilde{d}$	0.057** (0.021)	0.059** (0.022)
$1927 \times \tilde{d}$	0.026 (0.028)	0.029 (0.028)
$1928 \times \tilde{d}$	0.061* (0.030)	0.060* (0.030)
$1929 \times \tilde{d}$	0.046 (0.029)	0.043 (0.029)
$1930 \times \tilde{d}$	-0.018 (0.022)	-0.015 (0.022)
$1931 \times \tilde{d}$	0.004 (0.014)	0.008 (0.014)
$1933 \times \tilde{d}$	-0.068*** (0.013)	-0.063*** (0.013)
$1934 \times \tilde{d}$	-0.032*** (0.011)	-0.035*** (0.010)
$1935 \times \tilde{d}$	0.025** (0.010)	0.029*** (0.010)
$1936 \times \tilde{d}$	0.001 (0.012)	0.007 (0.012)
$1937 \times \tilde{d}$	0.015 (0.016)	0.011 (0.016)
$1938 \times \tilde{d}$	0.000 (0.019)	0.009 (0.019)
$1939 \times \tilde{d}$	-0.021 (0.015)	-0.021 (0.015)
$1940 \times \tilde{d}$	-0.005 (0.016)	-0.011 (0.016)
R^2	0.221	0.229
Observations	5,572	5,853

Note. The Round 1 sample inadvertently excluded 37 unexposed firms that lacked a prior-year observation during 1931–1935; the revised sample excludes only the 96 firms that retired their gold-clause debt during the litigation window. Standard errors in parentheses are two-way clustered by firm and year. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table R2: Table 4, column 7 (bank-debt placebo), before and after the clustering correction

	Round 1	Revised
Q	0.095*** (0.014)	0.095*** (0.017)
$\tilde{d}$	-0.007 (0.110)	-0.007 (0.093)
$1926 \times \tilde{d}$	-0.335* (0.177)	-0.335** (0.129)
$1927 \times \tilde{d}$	-0.210 (0.157)	-0.210 (0.124)
$1928 \times \tilde{d}$	-0.144* (0.078)	-0.144*** (0.038)
$1929 \times \tilde{d}$	0.252 (0.321)	0.252 (0.147)
$1930 \times \tilde{d}$	-0.097 (0.098)	-0.097 (0.081)
$1931 \times \tilde{d}$	-0.045 (0.110)	-0.045 (0.103)
$1933 \times \tilde{d}$	0.057 (0.093)	0.057 (0.061)
$1934 \times \tilde{d}$	0.060 (0.084)	0.060 (0.038)
$1935 \times \tilde{d}$	-0.026 (0.056)	-0.026 (0.026)
$1936 \times \tilde{d}$	-0.013 (0.062)	-0.013 (0.032)
$1937 \times \tilde{d}$	-0.038 (0.067)	-0.038 (0.041)
$1938 \times \tilde{d}$	0.025 (0.064)	0.025 (0.045)
$1939 \times \tilde{d}$	-0.056 (0.075)	-0.056 (0.052)
$1940 \times \tilde{d}$	0.032 (0.076)	0.032 (0.057)
R^2	0.239	0.239
Observations	7,074	7,074

Note. Coefficients are identical in the two versions. The Round 1 standard errors were inadvertently clustered by firm only; the Revised standard errors are two-way clustered by firm and year, like every other column of Table 4. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table R3: Internet Appendix Table IA.16 (columns 2–10 of Table 7 re-estimated with industry-year fixed effects)

	All controls	Q	Assets	NI/assets	Cash/assets	Payout	Book leverage	Market leverage	LT lia.
	linear	deciles	deciles	deciles	deciles	deciles	deciles	deciles	deciles
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Q	0.093*** (0.019)	0.092*** (0.019)	0.090*** (0.017)	0.092*** (0.018)	0.091*** (0.017)	0.092*** (0.018)	0.090*** (0.017)	0.091*** (0.018)	0.090*** (0.017)
$\tilde{d}$	−0.059** (0.022)	−0.062*** (0.019)	−0.061*** (0.019)	−0.058*** (0.019)	−0.063*** (0.020)	−0.065*** (0.020)	−0.060*** (0.018)	−0.060*** (0.019)	−0.061** (0.020)
$1933 \times \tilde{d}$	−0.028* (0.014)	−0.049** (0.019)	−0.056** (0.019)	−0.055** (0.019)	−0.028* (0.016)	−0.050*** (0.014)	−0.058*** (0.019)	−0.049** (0.018)	−0.043** (0.016)
$1934 \times \tilde{d}$	−0.012 (0.016)	−0.024 (0.015)	−0.031* (0.016)	−0.030 (0.025)	−0.024 (0.016)	−0.031* (0.015)	−0.017 (0.017)	−0.017 (0.016)	−0.018 (0.016)
After $\times \tilde{d}$	0.003 (0.018)	−0.004 (0.017)	−0.006 (0.017)	−0.002 (0.026)	−0.004 (0.016)	−0.010 (0.015)	−0.008 (0.020)	−0.004 (0.019)	−0.003 (0.018)
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	No	No	No	No	No	No	No	No	No
Industry-year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	6,990	7,015	7,015	7,015	7,015	7,015	7,015	7,015	7,015

Notes. This table replicates columns 2–10 of Table 7 replacing the year fixed effects with SIC2 industry-year fixed effects, so that all specifications share the same fixed-effect structure as column 1 of Table 7. The characteristic-interacted controls are identical to those in Table 7: column (1) includes all base-period firm characteristics interacted with the four period indicators as linear controls; columns (2)–(9) each include decile dummies for a single base-period characteristic interacted with the four period indicators. Base-period characteristics are measured in 1930, or the first available year thereafter. All regressions include firm and SIC2 industry-year fixed effects. All variables are winsorized at the 0.5% and 99.5% levels within each year. Standard errors in parentheses are two-way clustered by firm and year. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.